

Large Numbers Around Us

CHAPTER

01

Learning Outcomes...

- Large Numbers
- Reading and Writing Numbers
- Comparing Numbers
- Ascending and Descending Order
- Exact and Approximate Values
- Estimating Sums and Differences
- Patterns in Products
- Interesting Facts about Large Numbers

Introduction

In previous classes, we have added, subtracted, multiplied and divided numbers and also enjoyed different patterns of numbers. Now, we shall learn and explore more about numbers.

Large Numbers

Observe the given pattern :

Largest 1-digit number + 1 = Smallest 2-digit number

$$(9) + 1 = 10 \quad (\text{Ten})$$

Largest 2-digit number + 1 = Smallest 3-digit number

$$(99) + 1 = 100 \quad (\text{Hundred})$$

Largest 3-digit number + 1 = Smallest 4-digit number

$$(999) + 1 = 1,000 \quad (\text{Thousand})$$

Largest 4-digit number + 1 = Smallest 5-digit number

$$(9,999) + 1 = 10,000 \quad (\text{Ten Thousand})$$

Largest 5-digit number + 1 = Smallest 6-digit number

$$(99,999) + 1 = 1,00,000 \quad (\text{Lakh})$$

Largest 6-digit number + 1 = Smallest 7-digit number

$$(9,99,999) + 1 = 10,00,000 \quad (\text{Ten Lakh})$$

Largest 7-digit number + 1 = Smallest 8-digit number

$$(99,99,999) + 1 = 1,00,00,000 \quad (\text{Crore})$$

Largest 8-digit number + 1 = Smallest 9-digit number

$$(9,99,99,999) + 1 = 10,00,00,000 \quad (\text{Ten Crore})$$

Thus, we can write the smallest or largest number for any number of digits.

Reading and Writing Numbers

There are two systems of numeration used for reading and writing the numbers :

- The Indian System
- The International System (American System)

The names of large numbers are read differently in India as compared to the other parts of the world. This will become clear from the following example :

Numerals	1	5	4	3	6	5	9	8	0	1
Indian System	Arabs	Ten Crores	Crores	Ten Lakhs	Lakhs	Ten Thousands	Thousands	Hundreds	Tens	Ones
International System	Billions	Hundred Millions	Ten Millions	Millions	Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones

Here, the given numeral is 1543659801.

We can see that, in Indian System of Numeration, it is written as: 1,54,36,59,801 and can be read as, "One Arab fifty four crore thirty six lakh fifty nine thousand eight hundred one".

While in International System of Numeration, it is written as: 1,543,659,801 and can be read as, "One billion five hundred forty three million six hundred fifty nine thousand eight hundred one".

To place commas, we use following rules :

- In Indian system, first comma comes after three digits from the right and the next comma comes after every next two digits.
- In International system, comma comes after every three digits starting from the right.

From the above table, we can find that :

- 1 Lakh = 100 Thousands
- 10 Lakhs = 1 Million
- 1 Crore = 10 Millions
- 10 Crores = 100 Millions
- 1 Arab = 1 Billion

Expanded Form of Numbers

The expanded form of numbers is a way of writing the numbers to show the value of each digit based on its place value. It breaks down a number into its individual digits, each multiplied by its place value (Ones, tens, hundreds, etc).

For Example; The expanded form of 426438 is $4 \times 100000 + 2 \times 10000 + 6 \times 1000 + 4 \times 100 + 3 \times 10 + 8 \times 1$

ILLUSTRATIONS

1 Write the number names of each of the following numbers :

(i) 5,04,326 (ii) 38,96,405 (iii) 1,23,40,008

Soln.: (i) 5,04,326 in words is written as;
Five lakh four thousand three hundred twenty six.

(ii) 38,96,405 in words is written as;

Thirty eight lakh ninety six thousand four hundred five.

(iii) 1,23,40,008 in words is written as;

One crore twenty three lakh forty thousand eight.

2 Write the following numbers in expanded form.

(i) 9863402 (ii) 23400

Soln.: (i) Expanded form of 9863402 is :

$$9 \times 1000000 + 8 \times 100000 + 6 \times 10000$$

$$+ 3 \times 1000 + 4 \times 100 + 0 + 2 \times 1$$

(ii) Expanded form of 23400 is :

$$2 \times 10000 + 3 \times 1000 + 4 \times 100 + 0 + 0$$

3 Write the place value of the underlined digits in each of the following :

(i) 38695

(ii) 4020803

(iii) 326152

Soln.: (i) Place value of 8 in 38695 = 8000

(ii) Place value of 4 in 4020803 = 4000000

(iii) Place value of 1 in 326152 = 100

4 Write the following numerals in words in both Indian and International System of Numeration.

(i) 394261 (ii) 5430030

(iii) 19860005

Soln.:

	Numerals	Indian System of Numeration	International System of Numeration
(i)	394261	Three lakh ninety four thousand two hundred sixty one.	Three hundred ninety four thousand two hundred sixty one.
(ii)	5430030	Fifty four lakh thirty thousand thirty.	Five million four hundred thirty thousand thirty.
(iii)	19860005	One crore ninety eight lakh sixty thousand five.	Nineteen million eight hundred sixty thousand five.

5 Place the commas in each of the following using both Indian and International System of Numeration.

(i) 1232843 (ii) 920040 (iii) 86920605

Soln.:

	Numerals	Indian System of Numeration	International System of Numeration
(i)	1232843	12,32,843	1,232,843
(ii)	920040	9,20,040	920,040
(iii)	86920605	8,69,20,605	86,920,605

Comparing Numbers

For comparing two numbers, we have to remember the following points :

- If the number of digits in the given numbers is not same, then the number with more digits is greater.
- If the number of digits in the given numbers is same, then
 - (i) Compare the digits at the first place from the left. The number with greater digit is greater.
 - (ii) If the digits at the first place from the left is same, then compare the digits at the second place. The number with greater digit is greater.
 - (iii) If the digits are still not same, continue the process until you get unequal digits at the corresponding places.

Ascending and Descending Order

- Arrangement of numbers from the smallest to the largest is called ascending order.
- Arrangement of numbers from the largest to the smallest is called descending order.

ILLUSTRATIONS

6 Compare and write '<', '>' or '=':

- (i) 50 lakhs ____ 5 million
- (ii) 60 crores ____ 60 million
- (iii) 70 crore ____ 7 billion

Soln.:

- (i) Since, 10 lakhs = 1 million
 $\therefore$ 50 lakhs = 5 million
- (ii) Since, 10 crores = 100 million, therefore,
 60 crores = 600 million > 60 million.
 $\therefore$ 60 crores > 60 million
- (iii) Since, 1 Arab = 100 crore = 1 billion
 $\therefore$ 70 crore < 7 billion

7 Compare the following:

- (i) 562341720 and 46234120
- (ii) 39812465 and 39684521
- (iii) 840176321 and 840276125

Soln.:

- (i) 562341720 has nine digits and 46234120 has eight digits. Therefore, 562341720 is greater than 46234120.

- (ii) The given numbers have 8 digits, so we will directly compare their values.

The first two digits of the given numbers are same i.e., 39, so compare their third digits.

39812465 $\rightarrow$ third digit 8

39684521 $\rightarrow$ third digit 6

Since, $8 > 6$, 39812465 is greater than 39684521

- (iii) The given numbers have 9 digits, so we will directly compare their values.

The first three digits of the given numbers are same i.e., 840, so compare their fourth digits.

840176321 $\rightarrow$ fourth digit 1

840276125 $\rightarrow$ fourth digit 2

Since, $1 < 2$, 840176321 is smaller than 840276125.

8 Arrange the following numbers in ascending order : 7324891, 6123489, 7452103, 5891324

Soln.: All the given numbers have 7 digits, so we will directly compare their values.

Compare based on the first digits:

7324891 $\rightarrow$ starts with 7

6123489 $\rightarrow$ starts with 6

7452103 $\rightarrow$ starts with 7

5891324 $\rightarrow$ starts with 5

Since, $5 < 6 < 7$, clearly 5891324 is the smallest number and 6123489 is the only number greater than 5891324.

Compare second digits of 7324891 and 7452103 to find the greatest number.

7324891 $\rightarrow$ second digit 3

7452103 $\rightarrow$ second digit 4

Since, $3 < 4$, 7324891 is smaller than 7452103.

Thus, $5891324 < 6123489 < 7324891 < 7452103$

9 Arrange in descending order : 95128734, 96238651, 95128467, 93871234

Soln.: All the given numbers have 8 digits, so we will directly compare their values.

All numbers start with 9, compare their second digits:

95128734 $\rightarrow$ second digit 5

96238651 $\rightarrow$ second digit 6

95128467 $\rightarrow$ second digit 5

93871234 $\rightarrow$ second digit 3

Since, $6 > 5 > 3$, clearly 93871234 is the smallest number and 96238651 is the greatest number.

Since, the third, fourth and fifth digits of 95128734 and 95128467 are also same, so compare their sixth digits.

95128734 $\rightarrow$ sixth digit 7

95128467 $\rightarrow$ sixth digit 4

Since, $7 > 4$, 95128734 is greater than 95128467.

Thus, $96238651 > 95128734 > 95128467 > 93871234$

Exact and Approximate Values

An exact value is the value that is completely accurate and not rounded or estimated in any way while approximate value means finding a number that is close to the actual number usually by rounding up or rounding down. Here are real life situations, where concept of rounding up or rounding down is used:

- (i) In a party, there are 18 people and chairs come in sets of 10.
So, you round up to 20 chairs, so that everyone can sit comfortably.
- (ii) You have ₹58 and each toy cost ₹5.
So, you round down your amount to ₹55 so that you can buy 11 toys.

Nearest Neighbours

To find the nearest neighbours of large numbers by rounding off, you need to round a number to the nearest 10, 100, 1000, 10,000, etc depending on its size.

Follow the following rules for round up and down a number :

- If the digit immediately after the place value you are rounding is followed by 5, 6, 7, 8, or 9, round the number up.
- If the digit immediately after the place value you are rounding is followed by 0, 1, 2, 3, or 4, round the number down.
- Change the digits to the right of the rounding place to zeroes.

For example, nearest neighbours of the number 638794216 are shown in the table below:

Place Value	Rounded Value
Nearest thousand	638794000
Nearest ten thousand	638790000
Nearest lakh	638800000
Nearest ten lakh	639000000
Nearest crore	600000000

Estimating Sums and Differences

To estimate the result of an addition or subtraction:

1. Round off each large number to a common, place value (like nearest lakh or crore).
2. Perform the operation (addition or subtraction) using the rounded off numbers.

For example; estimate the sum and difference of 48671342 and 21385479

Round off the numbers to the nearest crore for easier calculation:

- 48671342 rounded up to 50000000
- 21385479 rounded down to 20000000

Estimated Sum = $50000000 + 20000000 = 70000000$

Estimated Difference = $50000000 - 20000000 = 30000000$

ILLUSTRATIONS

10 Write the five nearest neighbours for these numbers:

(i) 48732645 (ii) 12787356

Soln.: (i) Given number = 48732645

Nearest thousand	48733000
Nearest ten thousand	48730000
Nearest lakh	48700000
Nearest ten lakh	49000000
Nearest crore	50000000

(ii) Given number = 12787356

Nearest thousand	12787000
Nearest ten thousand	12790000

Nearest lakh	12800000
Nearest ten lakh	13000000
Nearest crore	10000000

11 Find the estimated sum and difference of 78913245 and 21345987.

Soln.: Round off the given numbers to the nearest crore, we get

78913245 round up to 80000000 and

21345987 round down to 20000000

Estimated sum = 100000000

Estimated difference = 60000000.

Patterns in Products

A pattern in product is a repeated or regular way in which numbers behave when multiplied. It shows a predictable change or sequence in the result of multiplication.

These patterns help us to understand multiplication better, identify shortcuts, and sometimes even predict results without full calculations.

One of the simplest patterns in multiplication occurs when we multiply a number by 10, 100, 1000, etc. When we do this, we just move the digits to the left and add zeroes at the end of the number.

ILLUSTRATIONS

12 Multiply: (i) 78×5 (ii) $2 \times 72 \times 50$

Soln.: (i) $78 \times \frac{10}{2} = 39 \times 10 = 390$

(ii) $2 \times 72 \times 50 = 2 \times 72 \times \frac{100}{2} = 7200$

13 Multiply the numbers below quickly.

(i) 50×76 (ii) 125×96 (iii) 250×60

Soln.: (i) $50 \times 76 = \frac{100}{2} \times 76 = 100 \times 38 = 3800$

(ii) $125 \times 96 = \frac{1000}{8} \times 96 = 1000 \times 12 = 12000$

(iii) $250 \times 60 = \frac{1000}{4} \times 60 = 1000 \times 15 = 15000$

Some Interesting Patterns

There are many interesting patterns that can help us to solve multiplication problems more easily.

For example :

(i) $1 \times 1 = 1$

$11 \times 11 = 121$

$111 \times 111 = 12321$

$1111 \times 1111 = 1234321$

$11111 \times 11111 = 123454321$

In the above pattern, the product forms a number where digits increase from 1 up to n , and then decrease back to 1.



(ii) $66 \times 61 = 4026$

$$666 \times 661 = 440226$$

$$6666 \times 6661 = 44402226$$

$$66666 \times 66661 = 4444022226$$

Note : In this pattern, occurrence of 0 and 6 are fix in each step but 2 and 4 depend on the number of times 6 occurs in 61, 661, 6661 and so on.

(iii) $3 \times 5 = 15$

$$33 \times 35 = 1155$$

$$333 \times 335 = 111555$$

$$3333 \times 3335 = 11115555$$

Note : In this pattern, occurrence of 1 and 5 depends on the number of times '3' occurs in 3, 33, 333 and so on.

(iv) $101 \times 101 = 10201$

$$102 \times 102 = 10404$$

$$103 \times 103 = 10609$$

$$104 \times 104 = 10816$$

Note : In this pattern, the hundred's digit is increasing by 2 each time and the last two digits in each result are always perfect squares.

(v) Triangle Pattern with 1 to 9:

$$1 \times 9 + 2 = 11$$

$$12 \times 9 + 3 = 111$$

$$123 \times 9 + 4 = 1111$$

$$1234 \times 9 + 5 = 11111$$

$$12345 \times 9 + 6 = 111111$$

Note : The result is always a string of 1's.

(vi) Special Mirror Products:

$$101 \times 101 = 10201$$

$$1001 \times 1001 = 1002001$$

$$10001 \times 10001 = 100020001$$

Note: The result has a 2 in the middle with mirror symmetry.

ILLUSTRATIONS

14 Observe the following pattern and complete it.

$$11 \times 11 = 121$$

$$101 \times 101 = 10201$$

$$10101 \times 10101 = 102030201$$

$$1010101 \times 1010101 = \underline{\hspace{2cm}}$$

$$101010101 \times 101010101 = \underline{\hspace{2cm}}$$

Soln.: $11 \times 11 = 121$

$$101 \times 101 = 10201$$

$$10101 \times 10101 = 102030201$$

$$1010101 \times 1010101 = \underline{1020304030201}$$

$$101010101 \times 101010101 = \underline{10203040504030201}$$

15 Is it possible for the product of a 2-digit number and a 3-digit number to be a 3-digit or a 4-digit number, or must it always result in a 4-digit number or a 5-digit number?

Soln.: To find the results, we need to check the products of the two smallest numbers and the two greatest numbers.

Smallest 2-digit number = 10, greatest 2-digit number = 99.

Smallest 3-digit number = 100, greatest 3-digit number = 999.

Smallest product = $10 \times 100 = 1000$, which is a 4-digit number.

Greatest product = $99 \times 999 = 98901$, which is a 5-digit number.

Hence, the product of a 2-digit number and a 3-digit number always results in a 4-digit or 5-digit number.

16 Can the product of a 4-digit number and a 2-digit number ever be a 5-digit number?

Why or why not?

Soln.: Smallest 2-digit number = 10

Smallest 4-digit number = 1000

Greatest 2-digit number = 99

Greatest 4-digit number = 9999

Therefore, smallest product = $10 \times 1000 = 10000$

Greatest product = $99 \times 9999 = 989901$

Hence, the product of a 4-digit number and a 2-digit number is always either a 5-digit number or a 6-digit number.

Note: When multiplying an m -digit number by an n -digit number, the product will always have either $(m + n - 1)$ digits or $(m + n)$ digits.

Interesting Facts about Large Numbers

Large numbers are numbers that are much greater than what we use in daily life. In mathematics, science, and even in real life, we often come across huge numbers — especially when talking about money, population, distances in space, or data in computers.

Real-Life Examples of Large Numbers are :

- Population of India: Over 1.4 billion $\rightarrow 1,40,00,00,000$
- Distance from Earth to Sun: 149,600,000 kilometres
- Number of stars in the universe: Around 1,000,000,000,000,000,000,000 (that's 24 zeros)

ILLUSTRATIONS

17 Let us assume a parking lot can hold 200 cars. Then, how many cars can 500 such parking lots hold? Can they fit 1 lakh cars?

Soln.: Cars in 500 parking lots = 200×500
 $= 1,00,000$

Yes, they can hold 1 lakh cars.

18 The Earth's circumference is about 40,000 km. If a person walks 20 km every

day, can he walk around the Earth in one lifetime?

Soln.: Distance to walk = 40,000 km

Daily walking distance = 20 km/day

Time required = $40,000 \div 20$

$= 2,000$ days

$= 2,000 \div 365 \approx 5.5$ years

Yes! A person can walk around the Earth in less than 6 years if he walks 20 km daily.

CONCEPT MAP

LARGE NUMBERS AROUND US

Expanded Form of Numbers

The expanded form of numbers is a way of writing the numbers to show the value of each digit based on its place value. It breaks down a number into its individual digits, each multiplied by its place value (Ones, tens, hundreds, etc).

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An exact value is the value that is completely accurate and not rounded or estimated in any way while approximate value means finding a number that is close to the actual number usually by rounding up or rounding down.

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For comparing two numbers, we have to remember the following points :

- If the number of digits in the given numbers is not same, then the number with more digits is greater.
- If the number of digits in the given numbers is same, then
 - (i) Compare the digits at the first place from the left. The number with greater digit is greater.
 - (ii) If the digits at the first place from the left is same, then compare the digits at the second place. The number with greater digit is greater.
 - (iii) If the digits are still not same, continue the process until you get unequal digits at the corresponding places.

Nearest Neighbours

To find the nearest neighbours of large numbers by rounding off, you need to round a number to the nearest 10, 100, 1000, 10,000 etc depending on its size.

Estimating Sums and Differences

To estimate the result of an addition or subtraction:

1. Round off each large number to a common, place value (like nearest lakh or crore).
2. Perform the operation (addition or subtraction) using the rounded off numbers.

Ascending and Descending Order

Arrangement of numbers from the smallest to the largest is called ascending order.

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Patterns in Products

A pattern in product is a repeated or regular way in which numbers behave when multiplied. It shows a predictable change or sequence in the result of multiplication.

Reading and Writing Numbers

Indian System

Arabs		Crores		Lakhs		Thousands		Ones	
Ten Arabs	Arabs	Ten Crores	Crores	Ten Lakhs	Lakhs	Ten Thousands	Thousands	Hundreds	Tens
								Ones	

International System

Billions			Millions			Thousands			Ones		
Hundred Billions	Ten Billions	Billions	Hundred Millions	Ten Millions	Millions	Hundred Thousands	Ten Thousands	Thousands	Hundreds	Tens	Ones

Solved Examples

1. Write the following numbers in the expanded form.

(i) 60,01,009

(ii) 8,56,43,210

Soln.: (i) In the expanded form, 60,01,009 is written as :

$$\begin{aligned} 60,01,009 &= 6 \times 10,00,000 + 0 \times 1,00,000 + 0 \times 10,000 \\ &\quad + 1 \times 1,000 + 0 \times 100 + 0 \times 10 + 9 \times 1 \\ &= 6 \times 10,00,000 + 1 \times 1,000 + 9 \end{aligned}$$

(ii) In the expanded form, 8,56,43,210 is written as:

$$\begin{aligned} 8,56,43,210 &= 8 \times 1,00,00,000 + 5 \times 10,00,000 \\ &\quad + 6 \times 1,00,000 + 4 \times 10,000 + 3 \times 1,000 \\ &\quad + 2 \times 100 + 1 \times 10 + 0 \times 1 \\ &= 8 \times 1,00,00,000 + 5 \times 10,00,000 + 6 \times 1,00,000 \\ &\quad + 4 \times 10,000 + 3 \times 1,000 + 2 \times 100 + 1 \times 10 \end{aligned}$$

2. The population of Karnataka, Delhi and Gujarat are respectively 54,37,211 ; 7,64,46,300 and 12,43,998. What is combined population of the three states? Also, write the total population of three states in International System of Numeration.

Soln.: Population of Karnataka is 54,37,211

Population of Delhi is 7,64,46,300

Population of Gujarat is 12,43,998

∴ Total population of three states = 54,37,211

7,64,46,300

+ 12,43,998

8,31,27,509

Total population (in International system)

= 83,127,509

In International System 83127509 can be written as: Eighty three million one hundred twenty seven thousand five hundred nine.

3. Write the following in words according to Indian System of Numeration :

(i) 5863496

(ii) 7425426

(iii) 25824320

(iv) 16289519

Soln.: (i) Fifty eight lakh sixty three thousand four hundred ninety six.

(ii) Seventy four lakh twenty five thousand four hundred twenty six.

(iii) Two crore fifty eight lakh twenty four thousand three hundred twenty.

(iv) One crore sixty two lakh eighty nine thousand five hundred nineteen.

4. Write the following in words according to International System of Numeration :

(i) 123527891

(ii) 183523512

(iii) 9058009

(iv) 2348828

Soln.: (i) One hundred twenty three million five hundred twenty seven thousand eight hundred ninety one.

(ii) One hundred eighty three million five hundred twenty three thousand five hundred twelve.

(iii) Nine million fifty eight thousand nine.

(iv) Two million three hundred forty eight thousand eight hundred twenty eight.

5. Write the numerals and place the commas.

(i) Seven crore thirty two lakh fifty four thousand two hundred sixty three.

(ii) One hundred eighty five million three hundred twenty three thousand two hundred fifty one.

(iii) Sixty two lakh nine thousand six hundred nineteen.

(iv) Two million three hundred forty six thousand eight hundred twenty five.

(v) One hundred million two hundred thousand three hundred.

Soln.: (i) 7,32,54,263

(ii) 185,323,251

(iii) 62,09,619

(iv) 2,346,825

(v) 100,200,300

6. Fill in the blanks :

(i) 9 crores = _____ million

(ii) 3 billion = _____ crores

(iii) 6 million = _____ lakhs

(iv) 80 crores = _____ million

Soln.: (i) Since, 1 crore = 10 million, therefore,
9 crores = $9 \times 10 = 90$ million

(ii) Since, 1 billion = 100 crores, therefore,
3 billion = $3 \times 100 = 300$ crores

(iii) Since, 1 million = 10 lakhs, therefore,
6 million = $6 \times 10 = 60$ lakhs

(iv) Since, 1 crore = 10 million, therefore,
80 crores = $80 \times 10 = 800$ million

7. Compare the following :

(i) 71245821 and 81245821

(ii) 65321847 and 65321846

(iii) 45789654 and 45321457

Soln.: (i) Both numbers have 8 digits.

Compare the first digits:

Since, $7 < 8$, therefore, $71245821 < 81245821$

(ii) Both numbers have 8-digit numbers and all the digit except the unit digit are same. So, compare their unit digit.

Since, $7 > 6$, therefore, $65321847 > 65321846$

(iii) The given numbers have 8 digits, so we will directly compare their values.

The first two digits of the given numbers are same i.e., 45. So, compare their third digits.

45789654 → third digit 7

45321457 → third digit 3

Since, $7 > 3$, $45789654 > 45321457$.

8. Arrange in descending order :

95851236, 95841235, 94841236, 96841236

Soln.: All the given numbers have 8 digits, so compare the digit-by-digit until the unequal corresponding digits:

First digits: 9, 9, 9, 9

Second digits: 5, 5, 4, 6

⇒ 96841236 is the largest number and 94841236 is the smallest number.

Since, second and third digits of 95851236 and 95841235 are same, so compare their fourth digit.

Fourth digit: $5 > 4$

Thus, $96841236 > 95851236 > 95841235 > 94841236$

9. Arrange in ascending order :

8420131, 7321582, 8321942, 7420156

Soln.: All the given numbers have 7 digits, so compare digit-by-digit until the last digit:

First digits: 8, 7, 8, 7

Compare the numbers with 7 as the first digit:
 $7321582 < 7420156$

Compare the numbers with 8 as the first digit:
 $8321942 < 8420131$

Thus, $7321582 < 7420156 < 8321942 < 8420131$

10. Round off to the nearest thousand, ten thousand, lakh, ten lakh, crore and ten crores :

(i) 165823478

(ii) 534892715

Soln.: (i) Given number 165823478

Nearest thousand	165823000
Nearest ten thousand	165820000
Nearest lakh	165800000
Nearest ten lakhs	166000000
Nearest crore	170000000
Nearest ten crores	200000000

(ii) Given number 534892715

Nearest thousand	534893000
Nearest ten thousand	534890000
Nearest lakh	534900000
Nearest ten lakhs	535000000
Nearest crore	530000000
Nearest ten crores	500000000

11. Estimate the sum and difference of 21947853 and 67438291.

Soln.: $21947853 \approx 22000000$

$67438291 \approx 67000000$

Estimated sum: $67000000 + 22000000 = 89000000$

Estimated difference: $67000000 - 22000000$
 $= 45000000$

12. Make the greatest and the smallest 9 - digit numbers using any five different digits with the condition given below:

(i) Digit 8 is always at ten lakhs place.

(ii) Digit 1 is always at crore place.

Soln.: The digits are 9, 8, 7, 6, 5, 4, 3, 2, 1, 0

(i) 5 Greatest digits are 9, 8, 7, 6, 5. Keeping digit 8 at ten lakhs place, the greatest number using different 5 digits is 998999765. Keeping 8 at ten lakhs place, other 4 smallest digits are 0, 1, 2, 3. So, the smallest number using 5 different digits is 108000023.

(ii) Keeping digit 1 at crore place, other 4 greatest digits are 9, 8, 7, 6 the greatest number using 5 different digits is 919999876. Keeping digit 1 at crore place, other 4 smallest digits are 0, 2, 3, 4. So, the smallest number using 5 different digits is 110000234.

13. Solve using the shortcuts:

(i) 92×25

(ii) $125 \times 12 \times 25 \times 8$

Soln.: (i) $92 \times 25 = 92 \times \frac{100}{4} = \frac{9200}{4} = 2300$

(ii) $125 \times 12 \times 25 \times 8$

$$= \frac{1000}{8} \times 12 \times \frac{100}{4} \times 8 = \frac{9600000}{32} = 300000.$$

14. Solve the following products quickly using multiplication shortcut method.

(i) 154×5

(ii) 1648×50

Soln.: (i) $154 \times 5 = 154 \times \frac{10}{2} = 77 \times 10 = 770$

(ii) $1648 \times 50 = 1648 \times \frac{100}{2} = 824 \times 100 = 82400$

15. Calculate the products and then describe the pattern you observe in the answers.

$3 \times 37 =$

$6 \times 37 =$

$9 \times 37 =$

$12 \times 37 =$

Soln.: $3 \times 37 = 111$

$6 \times 37 = 222$

$9 \times 37 = 333$

$12 \times 37 = 444$

Pattern: If you multiply $3n \times 37$, where $n = 1, 2, 3, 4, \dots$, then $3n \times 37 = 111 \times n$.

16. Calculate the products and then describe the pattern you observe in the answers.

$66 \times 66 =$

$666 \times 666 =$

$6666 \times 6666 =$

$66666 \times 66666 =$

Soln.: $66 \times 66 = 4356$

$666 \times 666 = 443556$

$6666 \times 6666 = 44435556$

$66666 \times 66666 = 4444355556$

Pattern: In this pattern, occurrence of 4 and 5 is one less than number of digit 6 occurs in number.

17. Perform the following multiplications. Do you notice any similarities or repeated number patterns in the answers?

$12345679 \times 9 =$

$12345679 \times 18 =$

$12345679 \times 27 =$

$12345679 \times 36 =$

Soln.: $12345679 \times 9 = 111,111,111$

$12345679 \times 18 = 222,222,222$

$12345679 \times 27 = 333,333,333$

$12345679 \times 36 = 444,444,444$

Pattern: In this pattern, multiplying 12345679 by $9n$ is the same as multiplying 111111111 by n .

18. Is it possible for a 4-digit number multiplied by a 3-digit number to result in an 8-digit number?

Soln.: Greatest 4-digit number = 9999, Greatest 3-digit number = 999, Smallest 4-digit number = 1000 and Smallest 3-digit number = 100.

Greatest Product = $9999 \times 999 = 9989001$, Smallest Product = $1000 \times 100 = 100000$.

Here, we can see that the product of a 4-digit number and a 3-digit number is always either a 6-digit number or a 7-digit number. Hence, the required product can't be an 8-digit number.

19. India is about 3,200 km long (North to South). If someone walks 10 km per day, can he cover the whole distance in a year?

Soln.: Time = $3,200 \div 10 = 320$ days.

Yes! A person could walk the length of India (North to South) in less than 1 year at 10 km/day.

20. Each library has 25,000 books. How many books do 4,000 such libraries have? Is it more than 10 crore books?

Soln.: Number of books in 4,000 libraries
 $= 25,000 \times 4,000 = 10,00,00,000$.

There are 10 crore books in 4,000 libraries, hence it is not more than but equal to 10 crore books.

NCERT Section

Figure it Out

(Page 3)

1. According to the 2011 Census, the population of the town of Chintamani was about 75,000. How much less than one lakh is 75,000?

2. The estimated population of Chintamani in the year 2024 is 1,06,000. How much more than one lakh is 1,06,000?

3. By how much did the population of Chintamani increase from 2011 to 2024?

Figure it Out

(Page 6 & 7)

For each number given below, write expressions for at least two different ways to obtain the number through button clicks. Think like Chitti and be creative.

- | | |
|------------|-----------|
| (a) 8300 | (b) 40629 |
| (c) 56354 | (d) 66666 |
| (e) 367813 | |

Figure it Out

(Page 7)

1. For the numbers in the previous exercise, find out how to get each number by making the smallest number of button clicks and write the expression.

2. Do you see any connection between each number and the corresponding smallest number of button clicks?

3. If you notice, the expressions for the least button clicks also give the Indian place value notation of the numbers. Think about why this is so.

Figure it Out

(Page 9)

1. Read the following numbers in Indian place value notation and write their number names in both the Indian and American systems:

- | | |
|---------------|----------------|
| (a) 4050678 | (b) 48121620 |
| (c) 20022002 | (d) 246813579 |
| (e) 345000543 | (f) 1020304050 |

2. Write the following numbers in Indian place value notation:

- One crore one lakh one thousand ten
- One billion one million one thousand one
- Ten crore twenty lakh thirty thousand forty
- Nine billion eighty million seven hundred thousand six hundred

3. Compare and write '<', '>' or '=':

- 30 thousand _____ 3 lakhs
- 500 lakhs _____ 5 million
- 800 thousand _____ 8 million
- 640 crore _____ 60 billion

Figure it Out

(Page 14)

1. Find quick ways to calculate these products:

- | | |
|--|---------------------|
| (a) $2 \times 1768 \times 50$ | (b) 72×125 |
| (c) $125 \times 40 \times 8 \times 25$ | |

2. Calculate these products quickly.

- $25 \times 12 = \underline{\hspace{2cm}}$
- $25 \times 240 = \underline{\hspace{2cm}}$
- $250 \times 120 = \underline{\hspace{2cm}}$
- $2500 \times 12 = \underline{\hspace{2cm}}$
- $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = 120000000$

Figure it Out

(Page 19-21)

1. Using all digits from 0 – 9 exactly once (the first digit cannot be 0) to create a 10-digit number, write the –

- Largest multiple of 5
- Smallest even number

2. The number 10,30,285 in words is Ten lakhs thirty thousand two hundred eighty five, which has 42 letters. Give a 7-digit number name which has the maximum number of letters.

3. Write a 9-digit number where exchanging any two digits results in a bigger number. How many such numbers exist?

4. Strike out 10 digits from the number 12345123451234512345 so that the remaining number is as large as possible.

5. The words 'zero' and 'one' share letters 'e' and 'o'. The words 'one' and 'two' share a letter 'o', and the words 'two' and 'three' also share a letter 't'. How far do you have to count to find two consecutive numbers which do not share an English letter in common?

6. Suppose you write down all the numbers 1, 2, 3, 4, ..., 9, 10, 11, ... The tenth digit you write is '1' and the eleventh digit is '0', as part of the number 10.

(a) What would the 1000th digit be? At which number would it occur?

(b) What number would contain the millionth digit?

(c) When would you have written the digit '5' for the 5000th time?

7. A calculator has only '+10,000' and '+100' buttons. Write an expression describing the number of buttons clicks to be made for the following numbers:

- (a) 20,800
- (b) 92,100
- (c) 1,20,500
- (d) 65,30,000
- (e) 70,25,700

8. How many lakhs make a billion?

9. You are given two sets of number cards numbered from 1 – 9. Place a number card in each box below to get the

- (a) largest possible sum
- (b) smallest possible difference of the two resulting numbers.



10. You are given some number cards; 4000, 13000, 300, 70000, 150000, 20, 5. Using the cards get as close as you can to the numbers below using any operation you want. Each card can be used only once for making a particular number.

(a) 1,10,000: Closest I could make is $4000 \times (20 + 5) + 13000 = 1,13,000$

(b) 2,00,000:

(c) 5,80,000:

(d) 12,45,000:

(e) 20,90,800:

11. Find out how many coins should be stacked to match the height of the Statue of Unity. Assume each coin is 1 mm thick.

12. Grey-headed albatrosses have a roughly 7-feet wide wingspan. They are known to migrate across several oceans. Albatrosses can cover about 900 – 1000 km in a day. One of the longest single trips recorded is about 12,000 km. How many days would such a trip take to cross the Pacific Ocean approximately?

13. A bar-tailed godwit holds the record for the longest recorded non-stop flight. It travelled 13,560 km from Alaska to Australia without stopping. Its journey started on 13 October 2022 and continued for about 11 days. Find out the approximate distance it covered every day. Find out the approximate distance it covered every hour.

14. Bald eagles are known to fly as high as 4500 – 6000 m above the ground level. Mount Everest is about 8850 m high. Aeroplanes can fly as high as 10,000 – 12,800 m. How many times bigger are these heights compared to Somu's building?

Exercise

Multiple Choice Questions

LEVEL - 1

Large Numbers

- The sum of largest 6-digit number and smallest 6-digit number is _____.
 (a) 100000
 (b) 1999999
 (c) 99999
 (d) 1099999 (NCERT Page 2) **Ev**
- The expanded form of 40,02,382 is
 (a) $4 \times 10,00,000 + 2 \times 1,00,000 + 3 \times 1,000 + 8 \times 10 + 2 \times 1$
 (b) $4 \times 10,00,000 + 2 \times 1,000 + 3 \times 100 + 8 \times 10 + 2 \times 1$
 (c) $4 \times 1,00,000 + 2 \times 10,000 + 3 \times 1,000 + 8 \times 1$
 (d) None of these (Page 6) **U**
- Which of the following number is written according to International System of Numeration?
 (a) 15,26,430
 (b) 1,526,430
 (c) 1,52,64,30
 (d) None of these (Page 8) **An**
- The number 6242530 is written in words as
 (a) Six million two hundred forty two thousand thirty.
 (b) Sixty two lakh forty two thousand five hundred thirty.
 (c) Sixty lakh forty thousand five hundred thirty.
 (d) Sixty two lakh forty thousand five hundred thirty. (Page 8)
- Which of the following numbers is the smallest?
 (a) Ten lakh three
 (b) One million five thousand
 (c) One million
 (d) One crore (Page 8) **An**
- One million = _____.
 (a) 1 lakh
 (b) 10 lakh
 (c) 100 lakh
 (d) 1 crore (Page 8)
- What is the value of 1 crore in the American system?
 (a) 1 million
 (b) 10 million
 (c) 100 million
 (d) 1 billion (Page 8)
- "Seventy-three crore forty six thousand eight hundred forty two" in numerals is _____.
 (a) 7346428
 (b) 73046248
 (c) 730046842
 (d) 730064824 (Page 8)
- How many zeroes are there in 10 lakhs?
 (a) 6
 (b) 7
 (c) 8
 (d) 4 (Page 8) **R**
- Which one is greater 940376421 or 940476421?
 (a) 940376421
 (b) 940476421
 (c) Both are equal
 (d) None of these (Page 9)
- The successor of 1 million is _____.
 (a) 2 millions
 (b) 1000001
 (c) 100001
 (d) 10001 **Ev**
- Place value of a digit in a number is _____.
 (a) Face value $\times$ value of the just preceding place it occupies in place value chart
 (b) Face value $\times$ value of the place it occupies in the place value chart
 (c) Face value
 (d) None of these **U**
- Find the sum of the place values of 7 in 43,726,787.
 (a) 700700
 (b) 707007
 (c) 700707
 (d) 70700 **Ev**

Exact and Approximate Values

14. Estimated difference of $(47551673 - 32451258)$ to the nearest lakhs is _____.

(a) 14000000
(b) 15000000
(c) 15100000
(d) 16000000

(Page 11) **Ev**

15. Find the estimated difference of $43793125 - 12576036$ to the nearest ten thousands.

(a) 31210000 (b) 3119000
(c) 31299000 (d) 3119800 (Page 11)

16. Round off 9,81,22,678 to the nearest lakh.

(a) 9,81,00,000
(b) 9,82,00,000
(c) 9,80,00,000
(d) 10,00,00,000

(Page 11)

17. 9416580 rounds off to the nearest thousands is _____.

(a) 9417000
(b) 9416000
(c) 9416600
(d) 9416500

(Page 11) **Ev**

18. 57824123 rounded to the nearest crore is:

(a) 60000000 (b) 57000000
(c) 50000000 (d) 58000000 (Page 11)

Patterns in Products and Facts about Large Numbers

19. What is the shortcut used to calculate 142×50 ?

(a) Dividing by 5 and multiplying by 10
(b) Dividing by 10 and multiplying by 2
(c) Dividing by 2 and multiplying by 100
(d) None of the above (Page 13) **U**

20. How would you quickly calculate 1024×125 ?

(a) Multiply 1024 by 50 and divide by 2
(b) Multiply 1024 by 100 and divide by 2
(c) Multiply 1024 by 1000 and divide by 8
(d) Multiply 1024 by 100 and add 25

(Page 13) **An**

21. Radhika used $256 \times 25 = 6400$. Which of the following expressions uses the same technique?

(a) $256 \times 4 \div 100$
(b) $256 \times 8 \div 1000$

(c) $256 \times 1000 \div 8$

(d) $256 \times 100 \div 4$

(Page 13)

22. Value of $125 \times 4 \times 8 \times 50$ is

(a) 10000
(b) 20000
(c) 100000
(d) 200000

(Page 13) **Ev**

23. $125 \times 64 =$

(a) 2000 (b) 4000
(c) 8000 (d) 16000 (Page 13)

24. Follow the pattern, $6 \times 6 = 36$

$$66 \times 6 = 396$$

$$666 \times 6 = 3996$$

The value of $6666 \times 6 =$

(a) 39966
(b) 399966
(c) 39996
(d) 339996

(Page 14) **An**

25. Using the pattern, $777 \times 7 = 5439$

$$7777 \times 7 = 54439$$

$$77777 \times 7 = 544439$$

The value of $777777 \times 7 =$

(a) 5444439
(b) 54444439
(c) 544444439
(d) 5444444439

(Page 14)

26. What is the number of digits in the product of two 3-digit numbers most likely to be?

(a) 2 or 3 (b) 3 or 4
(c) 4 or 5 (d) 5 or 6 (Page 15)

27. If you blink about 15 times per minute, then how many times do you blink in 24-hours?

(a) 14400
(b) 16000
(c) 18400
(d) 21600

(Page 16-18) **Ev**

28. If a tree absorbed 21 kg of CO_2 per year then how much does a forest of 1,000 trees absorb?

(a) 2100 kg
(b) 21000 kg
(c) 210000 kg
(d) 2100000 kg

(Page 16-18)

29. You type 2 words per second. Can you type a 100,000-word document in a 12-hour day?

- (a) Yes
- (b) No
- (c) Can't say
- (d) None of these

(Page 16-18) **Ev**

30. A book has 300 pages, and Ram reads 20 pages/day. Then, Ram finishes the book in

- (a) 5 days
- (b) 10 days
- (c) 12 days
- (d) 15 days

Page 16-18

LEVEL - 2

31. What is the difference between the greatest and the smallest number that can be formed using all the digits 1, 5, 6, 2, 0, 4, 3 (without repetition, each used only once)?

- (a) 5411972
- (b) 5421971
- (c) 5519754
- (d) 5543120

Ev

32. Using only the digits 0 and 1, what is the greatest 8-digit number that can be formed if both digits are used an equal number of times?

- (a) 11110000
- (b) 11001100
- (c) 10101010
- (d) 11100100

U

33. What is the difference between the largest 8-digit number and the largest 8-digit number with all distinct digits?

- (a) 98765431
- (b) 1234567
- (c) 12345678
- (d) 12345677

34. What is the difference between the place value and face value of 8 in the greatest number formed using the digits 6, 9, 4, 2, 8, 7, 3, 1 (each used only once)?

- (a) 79992
- (b) 7999992
- (c) 79902
- (d) 80000

35. What is the smallest 8-digit number that can be formed using any 4 different digits, where each digit can be repeated at most twice, and 5 must be in the ones place?

- (a) 11003335
- (b) 10012255
- (c) 11002255
- (d) 11022335

36. If $16502034 = 10000000 + n + 500000 + 2000 + 30 + 4$, then the value of n is _____.

- (a) 6
- (b) 6000000
- (c) 60000
- (d) 600000

Ev

37. The place value of 5 in 12537094 is the same as _____.

- (a) $500000 + 100094$
- (b) $500094 - 90$
- (c) $400094 + 90006$
- (d) $500094 - 94$

38. Sum of the place values of 3 in number 4783649 and 7427923 is

- (a) 3000
- (b) 3300
- (c) 3030
- (d) 3003

Ev

39. Find the difference in the place values of two fives in the numeral 21574513.

- (a) 500500
- (b) 599500
- (c) 499500
- (d) 499900

40. The place value of the digit 6 in the number 87576238 is _____.

- (a) 6200
- (b) 6
- (c) 6000
- (d) 1000

41. The largest eight-digit odd number that can be formed with the digits 0, 7, 3, 6, 4, 1, 2, 9 (without repetition) is

- (a) 97643210
- (b) 97642103
- (c) 97643201
- (d) 98764320

42. How many clicks of '+1000' are needed to reach 234000?

- (a) 23 clicks
- (b) 234 clicks
- (c) 2340 clicks
- (d) 23400 clicks

Ap

43. Which expression gives a result of 100000?

- (a) $125 \times 16 \times 100$
- (b) $25 \times 40 \times 100$
- (c) 100×1000
- (d) Both (b) and (c)

An

44. What is the value of $64 \times 125 \times 32 \times 25 \times 10 \times 5$?

- (a) 3200000
- (b) 32000000
- (c) 320000000
- (d) 3200000000

Ev

45. Which of the following must be true for a 4-digit $\times$ 2-digit product to be a 6-digit number?

- (a) 2-digit number < 10
- (b) 4-digit number > 5000
- (c) 2-digit number > 99
- (d) Product $> 99,999$

An

46. What is the smallest 2-digit number that can be multiplied with largest 3-digit number to get a 5-digit result?

- (a) 11 (b) 50
(c) 91 (d) 99

47. For the pattern: $11 \times 21 = 231$

$$11 \times 32 = 352$$

$$11 \times 43 = 473$$

The value of 11×54 is

- (a) 495 (b) 594
(c) 544 (d) 954

48. For the pattern: $9 \times 9 + 7 = 88$

$$98 \times 9 + 6 = 888$$

$$987 \times 9 + 5 = 8888$$

The value of $987654 \times 9 + 2 =$

- (a) 88888 (b) 888888
(c) 8888888 (d) 88888888

49. If a machine stamps 1 item every 0.5 seconds, how many items can it stamp in an hour?

- (a) 1800 (b) 2400
(c) 3600 (d) 7200

50. A fan spins 518,400 times in a day. What was its average spin rate per second?

- (a) 6 (b) 7
(c) 8 (d) 9

Ev

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

51. For 9120300, how many '+10000' and '+100' clicks are needed?

- (a) 912 clicks of '+10000' and 3 of '+100'
(b) 9120 clicks of '+10000' and 30 of '+100'
(c) 912 clicks of '+10000' and 30 of '+100'
(d) 9120 clicks of '+10000' and 3 of '+100'

Ap

52. Which of the following is the smallest 9-digit number using the digits from 0-9 exactly once and is a multiple of 10?

- (a) 102345678 (b) 102345679
(c) 123456790 (d) 123456780

U

53. Which of the following statements is/are incorrect?

(i) 70 million is equal to 7 crores.

(ii) 1 million is the smallest 8-digit number.

(iii) There are seven zeroes in 1 crore.

(iv) ₹64,78,45,329 rounded off to the nearest crore is ₹65,00,00,000

- (a) Only (ii) (b) Both (i) and (ii)
(c) Only (iii) (d) Both (ii) and (iv)

An

54. Vidhi writes the smallest 8-digit number, and Ridhi writes the greatest 9-digit number on the blackboard. Their Maths teacher finds the difference between the sum of the two numbers written and the smallest 5-digit number. What would be the answer?

- (a) 1000009999 (b) 1009989999
(c) 1099990001 (d) 1000000000

55. If a number is formed by interchanging the digits at the lakhs and ten thousands places of 834976219, then which of the following is correct?

- (a) New number > Original number
(b) New number < Original number
(c) New number = Original number
(d) None of these

Ap

56. Which of the following options is arranged in descending order?

- (a) 7,39,154; 7,93,154; 7,93,541; 7,93,951
(b) 8,50,76,745; 8,50,76,547; 8,50,67,574; 8,50,67,547
(c) 4,76,098; 4,87,678; 76,908; 87,876
(d) 3,15,45,001; 3,51,54,100; 4,15,45,001; 5,25,45,010

An

57. If $x \times 50 \times 4 \times 10 = 1,000,000$, then the value of x is

- (a) 500 (b) 1000
(c) 1250 (d) 2500

Ev

58. Which amongst the following products gives the largest 5-digit number?

- (a) 1500×66 (b) 3000×30
(c) 1999×49 (d) 9999×9

59. Using below pattern:

$$\begin{aligned} 1 \times 1 &= 1 \\ 11 \times 11 &= 121 \\ 111 \times 111 &= 12,321 \\ 1,111 \times 1,111 &= 1,234,321 \\ 11,111 \times 11,111 &= 123,454,321 \\ 111,111 \times 111,111 &= 12,345,654,321 \end{aligned}$$

The sum of all the digits of the product obtained in the next sequence is

- (a) 40 (b) 42
(c) 49 (d) 54



60. A fan spins at 3 spins/sec for 12 hours, then at 6 spins/sec for the next 12 hours. How many times does it spin in the whole day?

- (a) 345,600 (b) 518,400
(c) 777,600 (d) 388,800

Match the Following

In this section each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (a), (b), (c) and (d) out of which one is correct.

1. Match the following :

List-I

- (P) 1 lakh
(Q) 1 thousand
(R) 1 crore
(S) 10 lakh
(a) P-1, Q-2, R-3, S-4
(b) P-3, Q-1, R-2, S-4
(c) P-3, Q-1, R-4, S-2
(d) P-4, Q-2, R-3, S-1

List-II

1. 100 tens
2. 1 million
3. 100 thousands
4. 10 millions

2. Match the following :

List-I

- (P) Round off 2687 to the nearest 1000 is
(Q) Nine million forty six thousand one hundred fifty is
(R) $5 \times 1,00,000 + 2 \times 10,000 + 3 \times 1,000 + 6 \times 100 + 5$ is equal to
(S) Smallest 6-digit number using only one digit is

List-II

1. 111111
2. 5,23,605
3. 9046150
4. 3000

- (a) P-2, Q-1, R-3, S-4
(b) P-4, Q-3, R-2, S-1
(c) P-1, Q-2, R-3, S-4
(d) P-4, Q-2, R-3, S-1

3. Match the following:

List-I

- (P) Value of $32 \times 5 \times 25$ is _____
(Q) When a greatest 5-digit number is multiplied with a smallest 4-digit number, then we always get a _____ digit number.
(R) If Sunil drinks 1 glass of water in every 2 hours, then he drinks _____ glass of water in a day.
(S) If Yash reads 1 page in every 90 seconds, then in 12 hours he can read _____ pages.

List-II

1. 480
2. 12
3. 8
4. 4000

- (a) P-4, Q-3, R-1, S-2 (b) P-4, Q-2, R-3, S-1
(c) P-4, Q-3, R-2, S-1 (d) P-4, Q-2, R-1, S-3



Assertion & Reason Type

Directions : In each of the following questions, a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statements, mark the correct answer as

- (a) If both assertion and reason are true and reason is the correct explanation of assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of assertion.
(c) If assertion is true but reason is false.
(d) If assertion is false but reason is true.

1. **Assertion** : Smallest 7-digit number using only digit one is 1000000.

Reason : Smallest number is obtained by putting the smallest digit at the left most place except 0.

2. **Assertion** : The place value and face value of digit 8 in 18,002,403 is $8 \times 1,000,000$ and 8 respectively.

Reason : Place value is the value of digit itself and face value depends upon the place it occupies.

3. **Assertion** : After placing comma in the number 56943821 according to International System of Numeration, the number is written as 56,943,821.

Reason : To write a numeral in the Indian System, commas are first put after three digits from the right and then after every two digits.

4. **Assertion** : The product of 1600 and 25 can be evaluated as $1600 \times \frac{100}{4} = 40000$.

Reason : Multiplying a number by 100 and then dividing by 4 gives the same result as multiplying it by 25.

5. **Assertion** : When multiplying a 5-digit number by a 2-digit number, the product must have 7 digits.

Reason : When multiplying an x -digit number by a y -digit number, the product will always have either $(x + y - 1)$ digits or $(x + y)$ digit.

Hundred	Hundreds
Tens	Tens
Ones	Ones

- How is One Million written in the Indian number system?
(a) 10,00,000 (b) 1,00,000
(c) 1,00,00,000 (d) 10,00,00,000
- How many millions make 50 crores?
(a) 10 (b) 500
(c) 50 (d) 5
- 4,56,78,912 has at the lakh period.
(a) 78 (b) 56
(c) 4 (d) 912
- The smallest 8-digit number is equal to million.
(a) 1 (b) 10
(c) 1000 (d) 100

PASSAGE-II : Observe the following patterns:

$$1 \times 9 + 1 = 10$$

$$12 \times 9 + 2 = 110$$

$$123 \times 9 + 3 = 1110$$

$$1234 \times 9 + 4 = 11110$$

$$12345 \times 9 + 5 = 111110$$

Another pattern:

$$1 \times 1 = 1$$

$$11 \times 11 = 121$$

$$111 \times 111 = 12321$$

$$1111 \times 1111 = 1234321$$

$$11111 \times 11111 = 123454321$$

Now, answer the questions based on the patterns above.

- In the first pattern, what is the result of $123456 \times 9 + 6$?
(a) 1111110 (b) 1111116
(c) 1234566 (d) 111110
- What will be the result of 111111×111111 in the second pattern?
(a) 123456654321
(b) 11111111111
(c) 12345654321
(d) 12345678987654321

Comprehension Type

PASSAGE-I : Study the following table carefully and answer the questions :

Indian system of Numeration	International system of Numeration
Arab	Billions
Ten Crores	Hundred Million
Creore	Ten Millions
Ten Lakhs	Million
Lakh	Hundred Thousands
Ten Thousand	Ten Thousands
Thousand	Thousand

3. What is the logic behind the first pattern?
 - (a) Adding the number to its reverse
 - (b) Subtracting the number from a higher power of 10
 - (c) Multiplying the number by 10
 - (d) Multiplying an increasing sequence starting from 1 by 9 and adding the last digit
4. Predict the result of the expression $1234567 \times 9 + 7$ using the first pattern.
 - (a) 1234567 (b) 11111117
 - (c) 11111110 (d) 1111110



Subjective Problems

Very Short Answer Type

1. Express "Eight crore five thousand six" in numerals.
2. Rearrange the given numbers in ascending order.
5,90,672; 5,09,762; 2,69,705; 7,06,259; 9,07,526.
3. Find the sum of the place values of 7 and 4 in 43,627,897.
4. Write the smallest 8-digit number having five different digits.
5. Insert commas according to Indian System of Numeration :
(i) 28754263 (ii) 3864259
6. Compare the numbers 62347865 and 59864785.
7. Find the smallest 8-digit number formed by using digits 5, 0, 3, 2, 6, 1, 8, 9 each only once.
8. Evaluate 328×125 using multiplication shortcut method.
9. What is the smallest possible product of two 3-digit numbers?
10. If 120 cars are produced every hour in a factory, will 3,000 cars be produced in a day?

Short Answer Type

1. Write the number names of each of the following:
 - (i) 534,567,891
 - (ii) 23,46,891
 - (iii) 386,212
 - (iv) 2,12,42,861
2. Write the numerals with commas of each of the following :
 - (i) Six million fifty thousand three hundred six.
 - (ii) Twenty six lakh forty eight thousand two hundred five.
 - (iii) Eight hundred eight million eighty thousand eight hundred eight.
 - (iv) Two crore twenty lakh twenty two thousand two hundred two.
3. Write the following in expanded form.
 - (i) 286542 (ii) 19360654
 - (iii) 1212121
4. Write the following in standard form.
 - (i) $70,00,000 + 80,000 + 900$
 - (ii) $4,00,000 + 50,000 + 5,000 + 50 + 5$
 - (iii) $36,000 + 400 + 20 + 5$
5. Insert comma in the following numbers according to Indian and International System of Numeration and write number names.
 - (i) 321543302 (ii) 136843
6. Estimate the sum of 144578967 and 531812018 to the nearest crore.
7. Arrange in ascending order:
89,10,234; 80,10,234; 88,10,234; 84,10,234
8. Solve the following products quickly using multiplication shortcut method.
 - (i) 124×5
 - (ii) 240×25
9. Evaluate and identify the given pattern:

$$1 \times 19 + 1 =$$

$$12 \times 19 + 2 =$$

$$123 \times 19 + 3 =$$

$$1234 \times 19 + 4 =$$

10. What is the expected number of digits in the product of:
 - (i) 3-digit number and 4-digit number
 - (ii) 5-digit number and 1-digit number
11. If 45 emails are sent every 10 minutes, will 6,500 emails be sent in 24 hours?

Long Answer Type

1. Estimate the following sums or differences (nearest lakh) and then check whether the exact sum or difference is more than or less than your estimation.
 - (i) $4,95,980 + 4,44,100$
 - (ii) $9,17,899 - 1,65,401$
2. The Lazy Counter can press only +1, +10, +100, or +1000 buttons.
 - (i) How can it reach 4603 in two different ways?
 - (ii) How can it reach 17612 using three different combinations?
3. Use mental math to evaluate the following product.
 - (i) $4 \times 25 \times 16$
 - (ii) $16 \times 125 \times 5$
 - (iii) $50 \times 64 \times 125$
 - (iv) 12×250

4. Evaluate and identify the given pattern:

$$\begin{aligned}
 9 \times 9 + 7 &= \\
 98 \times 9 + 6 &= \\
 987 \times 9 + 5 &= \\
 9876 \times 9 + 4 &= \\
 98765 \times 9 + 3 &= \\
 987654 \times 9 + 2 &= \\
 9876543 \times 9 + 1 &=
 \end{aligned}$$

Integer/Numerical Value Type

1. What is the digit at the thousands place of 7,21,584?
2. Which digit of the number 871,253,564 has the same face and place value?
3. The face value of 7 in 4,26,76,254 is _____.

4. How many zeroes are there in ten lakh?
5. What is sum of digits of the greatest 8-digit number formed by two different digits?
6. Value of 400×125 using a suitable shortcut is _____.
7. The product of two 6-digit numbers can have a maximum of _____ digits.
8. If 15 students enter in a museum every 5 minutes, then _____ students enter in 2 hours.



Case Based Questions

Case I : Rohit, Arjun, Seema, Palak and Roshni were playing with some chits. In every chit, some numbers are written. They picked the chits one by one.



Based on the above information, answer the following questions.

CFQs

1. If Rohit picked the number 85236124, Arjun picked 52834456, Seema picked 82539741 and Roshni picked 38527258, then who picked the greatest number?
 - (a) Arjun
 - (b) Roshni
 - (c) Rohit
 - (d) Seema
2. If Palak picked 7580374, Seema picked 83024871, Rohit picked 93728456, Roshni picked 58263 and Arjun picked 629394, then which of the following is arranged in ascending order?
 - (a) Palak, Rohit, Roshni, Arjun, Seema
 - (b) Rohit, Seema, Arjun, Roshni, Palak
 - (c) Roshni, Seema, Palak, Arjun, Rohit
 - (d) Roshni, Arjun, Palak, Seema, Rohit
3. If Seema picked 3804672, Rohit picked 4567071, Arjun picked 1428753, Palak picked 9957242 and Roshni picked 2744321, then who picked the smallest number?
 - (a) Seema
 - (b) Arjun
 - (c) Rohit
 - (d) Palak

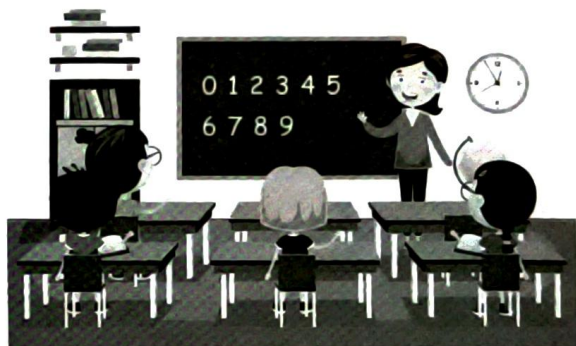
4. If Roshni picked 18346251, Arjun picked 2583416, Palak picked 3842178, Rohit picked 14521783 and Seema picked 9516436, then which of the following is arranged in descending order?

(a) Arjun, Palak, Seema, Rohit, Roshni
 (b) Roshni, Palak, Arjun, Rohit, Seema
 (c) Seema, Palak, Arjun, Roshni, Rohit
 (d) Roshni, Rohit, Seema, Palak, Arjun

5. If Arjun picked 2341178, Palak picked 1348321, Rohit picked 2178342, Roshni picked 1839753 and Seema picked 2454371, then who picked the second greatest number and the second smallest number?

(a) Arjun, Roshni
 (b) Seema Palak
 (c) Arjun, Palak
 (d) Rohit, Roshni

Case II : In a classroom activity, the teacher taught students about how to form numbers and then teacher wrote all the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9 on the board and asked some questions to the students.



Based on the above information, answer the following questions.

CFQs

- What is the smallest 8-digit number formed by the digits 0, 3, 8, 4, 5 and 1 using each digit atleast once?
 (a) 0113458 (b) 10003458
 (c) 10113458 (d) 10013458
- What is the greatest 6-digit even number formed by the digits 3, 4, 2, 0 and 1, repetition of digits is allowed?
 (a) 102344 (b) 443210
 (c) 432100 (d) 444444

3. What is the smallest 7-digit odd number formed by the digits 2, 3, 1, 6, and 5 using each digit atleast once?

(a) 1112365 (b) 1123561
 (c) 1235611 (d) 1111111

4. What is the smallest 6-digit odd number formed by the digits 3 and 8 using each digit atleast two times?

(a) 333883 (b) 338833
 (c) 333383 (d) 388883

Case III : Anny and John were two friends who lived in USA. Once they visited India. Anny knew about Indian System of Numeration but John didn't know. John asked some questions to Anny about Indian and International System of Numeration.



Based on the above information, answer the following questions.

CFQs

- How we write 'sixty eight million five hundred twenty seven thousand nine hundred one' in Indian System of Numeration?
 (a) Six crore eighty five lakh twenty seven hundred ninety one
 (b) Six crore eighty five lakh twenty seven thousand nine hundred one
 (c) Sixty eight lakh five hundred twenty seven thousand nine hundred one
 (d) Sixty eight crore five lakh twenty seven thousand nine hundred one
- How many crore(s) are there in ten million?
 (a) 1 (b) 10
 (c) 100 (d) none of these
- What is the correct number name for 96,461,306 in Indian System of Numeration?
 (a) Ninety six lakh four hundred sixty one thousand three hundred six
 (b) Ninety six crore forty six lakh one thousand three hundred six

- (c) Nine crore sixty four lakh sixty one thousand three hundred six
- (d) None of these

4. How can we write seventy two crore sixty eight lakh three hundred twenty two in International system?

- (a) Seventy two billion sixty two million three hundred twenty two
- (b) Seven hundred twenty six million eight hundred thousand three hundred twenty two
- (c) Seventy two million six hundred eighty thousand three hundred twenty two
- (d) None of these

5. What is the correct number name for 18220003 in Indian System of Numeration?

- (a) Eighteen crore twenty two lakh three
- (b) One crore eighty two lakh twenty thousand three
- (c) Eighteen crore two lakh twenty thousand three
- (d) One crore eight lakh twenty two thousand three

Case IV : The Government of India is planning to vaccinate the people of rural areas. For this they have appointed a Health Department, which is organising a vaccination camp for rural areas. 2,40,000 people need to be vaccinated. There are 40 mobile vans, and each can vaccinate 300 people per day.



Based on the above information, answer the following questions.

CFQs

1. How many people can be vaccinated by all 40 vans in one day?
 (a) 3,000 (b) 9,000
 (c) 12,000 (d) 15,000
2. How many total days are needed to vaccinate all 2,40,000 people?
 (a) 15 days (b) 20 days
 (c) 25 days (d) 30 days
3. If only 30 vans are used instead of 40, how many people can be vaccinated per day?
 (a) 9,000 (b) 6,000
 (c) 7,000 (d) 10,000
4. If the goal is to finish in 10 days, how many people must be vaccinated per day?
 (a) 12,000 (b) 15,000
 (c) 18,000 (d) 24,000
5. If each van operates for only 6 days, how many people will be vaccinated in total?
 (a) 72,000 (b) 84,000
 (c) 1,20,000 (d) 144,000